Groups 1 and 7, gases in the atmosphere and reactivity

30 questions: 10 multiple choice, 20 structured.

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1 · Part 1

Paper 1 & 2 Group 1 (alkali metals) – lithium, sodium and potassium [1]

Small pieces of lithium, sodium and potassium are added separately to water.

Which statement is true for all three metals?

A They sink and fizz slowly.

B They produce hydrogen gas and an alkaline solution.

C They burn with a lilac flame.

D They produce oxygen gas and an acidic solution.

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2 · Part 1

Paper 2 only Group 1 (alkali metals) – lithium, sodium and potassium [1]

The table shows the electronic configurations of atoms of three Group 1 elements, X, Y and Z.

Element	Electronic configuration
X	2.8.1
Y	2.8.8.1
Z	2.1

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Which shows the three elements in order of reactivity, most reactive first?

A Z, X, Y

B X, Y, Z

C X, Z, Y

D Y, X, Z

3 · Part 1

Paper 1 & 2 | Group 1 (alkali metals) – lithium, sodium and potassium | [6]

In a demonstration, a teacher adds a small piece of lithium to a large trough of water containing a few drops of universal indicator. She repeats the demonstration with sodium and then with potassium, using fresh water each time.

(a) State two observations that are the same for all three metals. [2]

(b) Complete the balanced equation for the reaction of sodium with water. Include state symbols.

....Na(s) + ...H_2O(l) → ..... + ... [2]

(c) Explain how these reactions with water provide evidence that lithium, sodium and potassium belong to the same family of elements. [2]

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4 · Part 1

Paper 1 & 2 Group 1 (alkali metals) – lithium, sodium and potassium [6]

Lithium, sodium and potassium are stored under oil. Tomás cuts a small piece from each metal and watches what happens to the freshly cut, shiny surface in the air.

Metal	What happens to the freshly cut surface in air
lithium	stays shiny for several minutes, then slowly turns dull
sodium	turns dull within about a minute
potassium	turns dull within a few seconds

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(a) Deduce the order of reactivity of the three metals, from most reactive to least reactive. Use evidence from the table to justify your answer. [2]

(b) Give a reason why these metals are stored under oil. [1]

(c) Lithium reacts with oxygen to form lithium oxide. Give the formula of lithium oxide. [1]

(d) Rubidium is below potassium in Group 1. Predict two observations when a small piece of rubidium is added to water. [2]

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5 · Part 1

Paper 1 & 2 | Group 1 (alkali metals) – lithium, sodium and potassium | [6]

The table gives some information about five Group 1 metals. The melting point of rubidium has been left out.

Metal	Melting point in °C	Density in g/cm^3
lithium	181	0.53
sodium	98	0.97
potassium	63	0.86
rubidium		1.53
caesium	28	1.93

(a) Describe the trend in melting point down Group 1. [1]

(b) Estimate the melting point of rubidium. [1]

(c) Predict the state of caesium on a day when the temperature is 35 °C. Give a reason for your answer. [2]

(d) A student says, "The density of the Group 1 metals increases down the group." Comment on whether the data in the table support this statement. [2]

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6 · Part 1

Paper 2 only

Group 1 (alkali metals) – lithium, sodium and potassium [5]

Potassium reacts much more vigorously with water than lithium does.

(a) Rubidium is in Group 1. State the number of electrons in the outer shell of a rubidium atom. [1]

(b) Explain, in terms of electronic configurations, why potassium is more reactive than lithium. [3]

(c) When sodium reacts with water, sodium atoms form sodium ions, Na^+. Give the electronic configuration of a sodium ion. [1]

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7 · Part 1

Paper 1 & 2 | Group 7 (halogens) – chlorine, bromine and iodine | [1]

Which row describes bromine at room temperature?

	Colour	Physical state
A	red-brown	liquid
B	yellow-green	gas
C	grey-black	solid
D	red-brown	gas

8 · Part 1

Paper 2 only | Group 7 (halogens) – chlorine, bromine and iodine | [1]

When a halogen reacts, each of its atoms gains one electron into its outer shell.

Which atom gains an electron most easily?

A an iodine atom

B an astatine atom

C a chlorine atom

D a bromine atom

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9 · Part 1

Paper 1 & 2 | Group 7 (halogens) – chlorine, bromine and iodine | [6]

Ngozi adds a few drops of aqueous solutions of three halogens to solutions of three potassium halides. She records the colour of each mixture. Chlorine water is very pale green, bromine water is orange and iodine solution is brown. The potassium halide solutions are colourless.

Halogen solution added	**potassium chloride solution**	**potassium bromide solution**	**potassium iodide solution**
chlorine water	not tested	orange	brown
bromine water	stays orange	not tested	X
iodine solution	stays brown	stays brown	not tested

(a) Predict the colour of the mixture at X. [1]

(b) Explain how the results show that the order of reactivity is chlorine > bromine > iodine. [3]

(c) Give a reason why Ngozi did not test chlorine water with potassium chloride solution. [1]

(d) Suggest one reason why Ngozi uses aqueous solutions of the halogens rather than the pure elements. [1]

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10 · Part 1

Paper 1 & 2 | Group 7 (halogens) – chlorine, bromine and iodine | [6]

The table gives the melting points and boiling points of four halogens. The values for astatine, which is below iodine in Group 7, have been left out.

Halogen	Melting point in °C	Boiling point in °C
fluorine	−220	−188
chlorine	−101	−34
bromine	−7	59
iodine	114	184
astatine		

(a) Describe the trend in boiling point down Group 7. [1]

(b) Deduce the physical state of bromine at 70 °C. Give a reason for your answer. [2]

(c) Predict the physical state and the colour of astatine at room temperature. [2]

(d) Estimate the boiling point of astatine. [1]

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11 · Part 1

Paper 2 only

Group 7 (halogens) – chlorine, bromine and iodine

[6]

Chlorine is more reactive than bromine.

(a) Give the electronic configuration of a chlorine atom. Use the Periodic Table. [1]

(b) Explain, in terms of electronic configurations, why chlorine is more reactive than bromine. [3]

(c) Iodine solution is added to a colourless solution of sodium astatide. Predict whether a reaction takes place. Explain your answer. [2]

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12 · Part 1

Paper 1 & 2 | Group 7 (halogens) – chlorine, bromine and iodine | [5]

Chlorine water is added to sodium iodide solution. The equation for the reaction is shown.

$Cl_2(aq) + 2NaI(aq) \rightarrow 2NaCl(aq) + I_2(aq)$

(a) State what is meant by oxidation, in terms of electrons. [1]

(b) Explain, in terms of electrons, why this reaction is a redox reaction. [2]

(c) Identify the oxidising agent in this reaction. Give a reason for your answer. [2]

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13 · Part 1 Paper 1 & 2 Gases in the atmosphere [1]

Which row shows the approximate percentages by volume of four gases in dry air?

	Nitrogen in %	Oxygen in %	Argon in %	Carbon dioxide in %
A	78	21	0.9	0.04
B	78	21	0.04	0.9
C	78	16	0.9	4
D	21	78	0.9	0.04

14 · Part 1 Paper 1 & 2 Gases in the atmosphere [1]

Which equation represents the thermal decomposition of copper(II) carbonate?

A $2CuCO_3(s) \rightarrow 2Cu(s) + 2CO_2(g) + O_2(g)$

B $CuCO_3(s) \rightarrow CuO_2(s) + CO(g)$

C $CuCO_3(s) \rightarrow CuO(s) + CO_2(g)$

D $CuCO_3(s) + 2HCl(aq) \rightarrow CuCl_2(aq) + H_2O(l) + CO_2(g)$

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15 · Part 1 Paper 1 & 2 Gases in the atmosphere [1]

Sulfur burns in oxygen.

Which equation, including state symbols, is correct for this reaction?

A $S(s) + O(g) \rightarrow SO(g)$

B $2S(s) + 3O_2(g) \rightarrow 2SO_3(g)$

C $S(s) + O_2(g) \rightarrow SO_2(s)$

D $S(s) + O_2(g) \rightarrow SO_2(g)$

16 · Part 1

Paper 1 & 2 | Gases in the atmosphere | [7]

Ayesha uses damp iron wool to find the percentage by volume of oxygen in air.

She pushes the damp iron wool into the closed end of a graduated tube. She stands the tube upside down in a beaker of water so that some air is trapped above the water. She reads the volume of trapped air at the start. After one week the water level has stopped changing, and she reads the volume again.

Reading	Volume of trapped air in cm^3
at the start	50.0
after one week	39.5

(a) Explain why the water rises up the tube during the week. [2]

(b) Calculate the percentage by volume of oxygen in the air. [2]

(c) Give a reason why Ayesha waits until the water level has stopped changing before taking the final reading. [1]

(d) Suggest why iron wool is used rather than a single iron nail. [1]

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(e) Ayesha repeats the experiment twice more. Her results are 20.6% and 20.8%. Calculate the mean of her three results. [1]

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17 · Part 1

Paper 1 & 2 Gases in the atmosphere [7]

Bilal measures the percentage by volume of oxygen in air using copper.

He connects two gas syringes to the two ends of a silica tube containing copper powder. One syringe contains 90.0 cm^3 of air and the other is empty. He heats the copper strongly and pushes the air backwards and forwards over the hot copper. Every two minutes he stops heating, lets the apparatus cool to room temperature, and records the total volume of gas.

Time of heating in minutes	**0**	**2**	**4**	**6**	**8**	**10**
Total volume of gas in cm^3	90.0	80.5	75.0	72.4	71.2	71.2

(a) Explain why Bilal pushes the air backwards and forwards over the copper many times. [1]

(b) Calculate the percentage by volume of oxygen in the air using Bilal's results. [2]

(c) Explain why Bilal lets the apparatus cool to room temperature before each reading. [2]

(d) Another student uses only a very small amount of copper powder. Explain how this would affect her value for the percentage of oxygen. [2]

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18 · Part 1 Paper 1 & 2 Gases in the atmosphere [6]

Magnesium ribbon and hydrogen gas both burn in oxygen.

(a) Describe what is seen when magnesium burns in oxygen. [2]

(b) Complete the balanced equation for the combustion of magnesium. Include state symbols.

....Mg(s) + O_2(g) → ..... [2]

(c) Hydrogen burns in oxygen with a pale blue flame. State the name of the product. Explain why this reaction is an oxidation. [2]

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19 · Part 1

Paper 1 & 2 Gases in the atmosphere [5]

Kofi heats 1.0 g of copper(II) carbonate in a test tube. A delivery tube carries the gas produced into a test tube of limewater. He records the time taken for the limewater to turn milky. He repeats the experiment with three other metal carbonates.

Carbonate	Time for the limewater to turn milky in s
copper(II) carbonate	25
zinc carbonate	55
magnesium carbonate	140
sodium carbonate	no change after 600

(a) Describe the colour change of the solid when copper(II) carbonate is heated. [1]

(b) List the four carbonates in order, starting with the one that decomposes most easily. [1]

(c) State two variables that Kofi should keep the same in each experiment. [2]

(d) Explain why Kofi lifts the delivery tube out of the limewater before he stops heating. [1]

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20 · Part 1

Paper 1 & 2 | Gases in the atmosphere | [6]

The table shows the concentration of carbon dioxide in the atmosphere, measured at the same observatory, in four different years. The concentration is given in parts per million (ppm).

Year	Concentration of carbon dioxide in ppm
1960	317
1980	339
2000	370
2020	414

(a) Describe how the concentration of carbon dioxide changed between 1960 and 2020. [2]

(b) Calculate the percentage increase in the concentration of carbon dioxide from 1960 to 2020. [2]

(c) Explain how increasing amounts of carbon dioxide in the atmosphere may contribute to climate change. [2]

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21 · Part 1 Paper 1 & 2 Reactivity series [1]

Which list shows four metals in order of decreasing reactivity, most reactive first?

A sodium, potassium, calcium, iron

B potassium, magnesium, zinc, copper

C magnesium, aluminium, iron, zinc

D gold, silver, copper, iron

22 · Part 1 Paper 1 & 2 Reactivity series [1]

Which mixture would react?

A copper and zinc sulfate solution

B iron and magnesium sulfate solution

C silver and copper(II) nitrate solution

D magnesium and iron(II) sulfate solution

23 · Part 1 Paper 1 & 2 Reactivity series [1]

Aluminium reacts with iron(III) oxide when the mixture is heated.

$2Al(s) + Fe_2O_3(s) \rightarrow Al_2O_3(s) + 2Fe(s)$

Which statement about this reaction is correct?

A Aluminium is oxidised because it loses electrons.

B Aluminium is reduced because it gains oxygen.

C Iron(III) oxide is the reducing agent.

D Iron(III) oxide is oxidised because it loses oxygen.

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24 · Part 1

Paper 1 & 2 | Reactivity series | [8]

Leila investigates the reactions of magnesium, zinc and iron with dilute hydrochloric acid.

(a) The table shows Leila's results. Deduce the order of reactivity of the three metals, from most reactive to least reactive. Use evidence from the table to justify your answer.

Metal	Observations	Temperature rise in °C
magnesium	vigorous fizzing; the metal disappears quickly	14.5
zinc	steady fizzing	4.0
iron	a few bubbles form slowly	1.5

[2]

(b) Complete the balanced equation for the reaction of zinc with dilute hydrochloric acid. Include state symbols.

Zn(s) + ...HCl(aq) → ..... + ... [2]

(c) Describe how Leila could carry out her investigation so that the comparison of the three metals is fair. [4]

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25 · Part 1 Paper 1 & 2 Reactivity series [6]

Some information about the reactions of three metals with water is given.

- Calcium reacts steadily with cold water, giving a stream of bubbles.
- Magnesium reacts very slowly with cold water, but reacts quickly when heated in steam, forming magnesium oxide and a gas.
- Copper does not react with water or with steam.

(a) State the names of the two products formed when calcium reacts with cold water. [2]

(b) Give the balanced equation for the reaction of magnesium with steam. Include state symbols. [2]

(c) Deduce the order of reactivity of calcium, copper and magnesium, from most reactive to least reactive. Explain your answer using the information given. [2]

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26 · Part 1

Paper 1 & 2 Reactivity series [7]

Hana adds pieces of three metals, R, T and W, to solutions of the nitrates of the same metals.

Metal added	solution of R nitrate	solution of T nitrate	solution of W nitrate
R	not tested	no reaction	reaction
T	reaction	not tested	reaction
W	no reaction	no reaction	not tested

(a) Deduce the order of reactivity of R, T and W, from most reactive to least reactive. Explain your answer. [2]

(b) Hana then adds a piece of zinc to blue copper(II) sulfate solution. State two observations she would make. [2]

(c) Complete the equation for the reaction of zinc with copper(II) sulfate solution. Include state symbols.

$Zn(s) + CuSO_4(aq) \rightarrow$ ..... + ... [1]

(d) Explain, in terms of electrons, why the reaction of zinc with copper(II) sulfate solution is a redox reaction. [2]

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27 · Part 1

Paper 1 & 2 | Reactivity series | [6]

Darius heats four mixtures of a metal powder with a metal oxide powder. He records his observations.

Mixture heated	Observation
magnesium and copper(II) oxide	the mixture glows brightly
copper and magnesium oxide	no change
zinc and copper(II) oxide	the mixture glows
copper and zinc oxide	no change

(a) Complete the balanced equation for the reaction of magnesium with copper(II) oxide. Include state symbols.

$Mg(s) + CuO(s) \rightarrow$ ..... + ... [2]

(b) Identify the substance that is reduced in the reaction between magnesium and copper(II) oxide. Give a reason for your answer in terms of oxygen. [2]

(c) Predict whether a reaction would take place when a mixture of iron powder and zinc oxide is heated. Explain your answer. [2]

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28 · Part 1

Paper 1 & 2 | Reactivity series | [6]

Ravi sets up four test tubes, each containing one clean iron nail. He leaves them for one week.

Tube	Contents of the tube	Result after one week
1	nail half covered by tap water; tube open to the air	some rust
2	nail covered by water that has been boiled and cooled, with a layer of oil on top of the water; tube stoppered	no rust
3	nail in air, with some anhydrous calcium chloride in the bottom of the tube; tube stoppered	no rust
4	nail half covered by salt water; tube open to the air	a lot of rust

(a) Explain why the water in tube 2 was boiled and then covered with a layer of oil. [2]

(b) Suggest the purpose of the anhydrous calcium chloride in tube 3. [1]

(c) Deduce the two substances that must both be present for iron to rust. Explain your answer using the results for tubes 1, 2 and 3. [2]

(d) State the effect of salt on the rusting of iron, using the results. [1]

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29 · Part 1

Paper 1 & 2 | Reactivity series | [8]

Iron and steel objects can be protected from rusting in different ways.

(a) Bicycle frames are often painted. Explain how painting prevents a bicycle frame from rusting. [1]

(b) Galvanised steel buckets are coated with a layer of zinc. Explain how the zinc protects the steel from rusting, both while the coating is complete and after the coating has been scratched so that some steel is exposed. [4]

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(c) Blocks of magnesium are attached to the steel hulls of some ships. Explain why these blocks must be replaced from time to time. [1]

(d) Suggest why attaching blocks of copper to a steel hull would not protect the hull from rusting. [2]

30 · Part 1 Paper 1 & 2 Reactivity series [7]

Metal M is not one of the metals in the reactivity series that you have studied. A student finds out the following about M.

- M does not react with cold water.
- M reacts slowly with dilute hydrochloric acid, giving a few bubbles of gas.
- M displaces copper from copper(II) sulfate solution.
- M does not displace iron from iron(II) sulfate solution.

(a) Deduce the position of M in the reactivity series. Explain your answer using the information given. [3]

(b) M forms ions with a 2+ charge, M^{2+}. Give the formula of the sulfate of M and the formula of the chloride of M. [2]

(c) Predict whether M would react when heated with copper(II) oxide. Explain your answer. [2]

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